Final Project

PSTAT122: Design and Analysis of Experiments

Winter 2026

STUDENT NAME

- Zachary Scholnick (3374824)
- Anjali Dev (A167A78)
- Logan Marks (9536947)

Due Date

Due Date: Monday, March 16, 2026, 11:59 PM

1 Introduction

In the field of map-making and cartography, scale is one of the most critical and challenging decisions practitioners face. At too large of a scale (zoomed in), key features are easily visible, but lose out on important context. At the same time, a very low scale (zoomed out), means that the information is difficult to see and fine spatial details are lost. Getting the scale of your map right is critical in conveying key information - and for that reason it's a question asked every time a map is created. On top of the practical aspects of mapmaking, humans possess a number of cognitive biases that make interpretations of proportion and area very difficult. As such, this experiment was motivated by the desire to provide real insights into what scale of representation makes estimation of area most intuitive.

While tackling this in entirety is beyond the scope of this project, we chose to focus on testing a key component of this issue: How does the scale of a mapped polygon affect the accuracy in which users are able to estimate the area? Do people systematically overestimate or underestimate polygon area at different scales?

2 Experimental Design

To assess this, we created 3 polygons of equal area, 1.71 square miles, selected because mapped parcels, agricultural units, and other targets of interest are commonly in the 1-3 square mile range. We also opted to map polygons of equal area to avoid introducing an additional confounding variable. These 3 polygons represented common archetypes in geometric representation: the simple polygon, the rounded polygon, and the more complex, concave polygon. We selected 3 random regions in the Patagonian Steppe in Argentina, as this region lacked easily recognizable reference points of scale, like trees or streams, and consisted mainly of rock formations of unidentifiable size. To ensure consistent areal calculation of mapped region and polygon, we opted for the POSGAR 2007/EPSG:5343 projection, which is the official geodetic reference system of Argentina.

Each polygon was then plotted at the scale ratios of 1:13,390, 1:26,780, and 1:59,852 to provide a large-scale, medium-scale, and small-scale plot. In a more generalizable interpretation, this means that we plotted polygons such that they covered 40%, 10%, and 2% of the entire mapped region. The scale bar itself was dynamically recalculated using QGIS based on the current map extent with a fixed display size for a constant visual across maps.

Table 1: Scale treatments and polygon coverage of map frame

Scale Label	Scale Ratio	Polygon Coverage
Large (L)	1:13,390	40%
Medium (M)	1:26,780	10%
Small (S)	1:59,852	2%

Since our project was trying to capture natural, intuitive estimates of area, participants were each shown a unique combination of scale and polygon type to prevent prior exposure from influencing estimates. Each participant receieved 3 treatment assignments, randomized using a Latin square structure to ensure that each participant only saw each scale level and polygon type exactly once in varying orders.

We expected that significant variation in spatial cognitive ability across tested individuals would be a major driver in estimation accuracy. By treating participants as a blocking factor, we can account for the inter-participant variability separate from the residual error, and boost our statistical power. We also hypothesized that polygon shape would alter estimation difficulty regardless of scale. By introducing 3 archetypal polygons as a fixed effect, we can control for this variability to better isolate the effect of scale.

Table 2: Example Latin square treatment assignment for one replicate block

	Curved Polygon	Simple Polygon	Concave Polygon
Participant 7	S	M	L
Participant 12	M	L	S
Participant 2	L	S	M

In this experiment, each participant was given 30 seconds per map to write down an area estimate in square miles of a curved, simple and concave polygon, each shown at a unique scale. This response was then transformed into a signed percent error:

$$\text{Percent Error} = \frac{\hat{a} - a}{a} \times 100$$

where $\hat{a}$ is the participant's estimate and $a = 1.71$ square miles is the true polygon area.

To formalize our experiment into a statistical model for analysis, we implemented a two-way ANOVA with blocking on participant. The response variable Y_{ijk} denotes the signed percent error for participant k receiving scale treatment i and polygon type j. The model is:

$$Y_{ijk} = \mu + \tau_i + \beta_j + \rho_k + \epsilon_{ijk}$$

- $i = 1, 2, 3$ indexes scale treatment (Large, Medium, Small)
- $j = 1, 2, 3$ indexes polygon type (simple, curved, concave)
- $k = 1, \cdots, 15$ indexes participant block
- μ is the overall mean percent error
- τ_i is the fixed effect of scale treatment i, with $\sum \tau_i = 0$
- β_j is the fixed effect of polygon type j, with $\sum \beta_j = 0$
- ρ_k is the fixed effect of participant block k, with $\sum \rho_k = 0$
- $\epsilon_{ijk} \sim \mathcal{N}(0, \sigma^2)$ is the residual error

3 Data Collection

Procedure Each participant was given 30 seconds to estimate the area in square miles of each polygon shown. Trials were conducted using a slide presentation displayed on a laptop screen, with each slide showing one polygon at its assigned scale alongside a scale bar. Maps were shown on the same monitor at the same display size to keep stimuli constant. Participants recorded their estimates on paper. All 5 replicates were conducted in the UCSB library on March 14, 2026. Participants were seated individually and shown the slides in sequence according to the generated Latin Squares, with no prior knowledge of the true polygon area or that all three polygons were equal in size.

Challenges/Adjustments Several challenges arose during data collection. The most notable was that some participants were unfamiliar with square miles as a unit of measurement, which may have introduced systematic bias into estimates regardless of scale treatment. While participants were not coached on how to use the scale bar, some may have relied on intuition rather than calculation. Additionally, the 30-second time limit, while intentional in capturing instinctive responses, may have prevented participants from making more deliberate estimates — meaning results may reflect first impressions of area rather than considered judgments.

Finally, as trials were conducted in a shared library space, there is a very small possibility that some participants overheard the estimates of others, which could have anchored subsequent guesses. Protocol was updated so that participants could not use outside props like a pencil, phone, or keys to match distances from the scale bar onto the map. This was to ensure constant conditions in estimating area across trials.

Data Presentation

Table 3: Full experimental data with percent error by participant, polygon type, and scale

	participant	replicate	row	col	scales	polygon	percent_error
1	P1	1	1	1	M	curved	180.70
2	P1	1	1	2	L	simple	163.16
3	P1	1	1	3	S	concave	75.44
19	P2	3	1	1	M	concave	75.44
20	P2	3	1	2	L	simple	75.44
21	P2	3	1	3	S	curved	75.44
40	P3	5	2	1	M	curved	16.96
41	P3	5	2	2	L	simple	-41.52
42	P3	5	2	3	S	concave	87.13
16	P4	2	3	1	M	curved	75.44
17	P4	2	3	2	L	concave	133.92
18	P4	2	3	3	S	simple	163.16
4	P5	1	2	1	S	curved	81.29
5	P5	1	2	2	M	simple	63.74
6	P5	1	2	3	L	concave	52.05
43	P6	5	3	1	S	curved	40.35
44	P6	5	3	2	M	simple	28.65
45	P6	5	3	3	L	concave	-6.43
31	P7	4	2	1	S	simple	192.40
32	P7	4	2	2	M	concave	601.75
33	P7	4	2	3	L	curved	250.88
37	P8	5	1	1	L	curved	40.35
38	P8	5	1	2	S	simple	46.20
39	P8	5	1	3	M	concave	16.96
10	P9	2	1	1	L	curved	-16.37
11	P9	2	1	2	S	concave	63.74
12	P9	2	1	3	M	simple	87.13
13	P10	2	2	1	S	curved	87.13
14	P10	2	2	2	M	concave	104.68
15	P10	2	2	3	L	simple	139.77
28	P11	4	1	1	M	simple	57.89

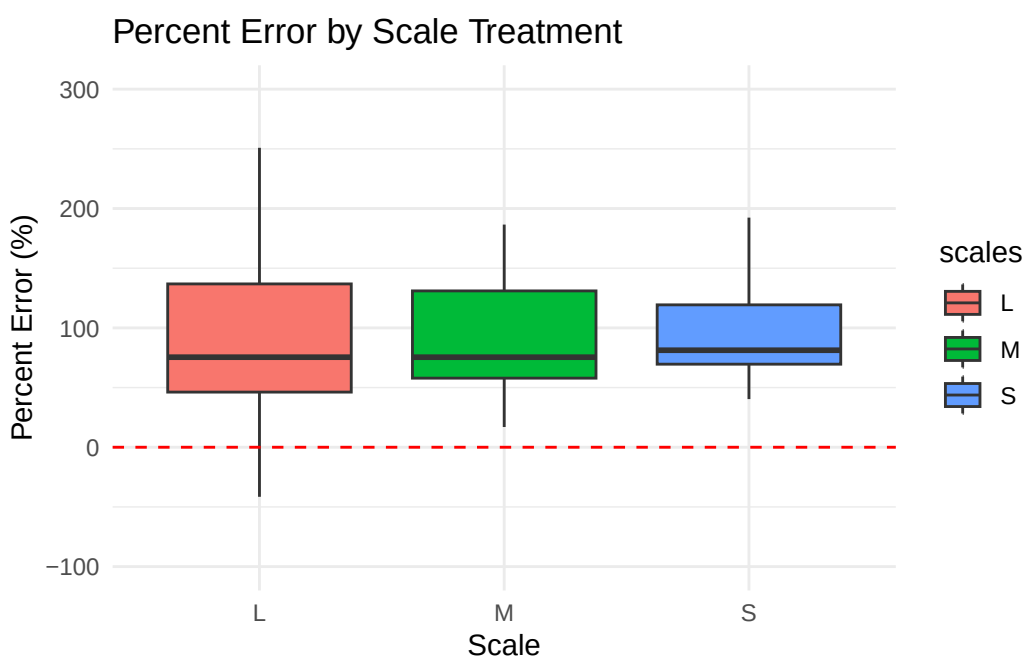
	participant	replicate	row	col	scales	polygon	percent_error
29	P11	4	1	2	L	concave	69.59
30	P11	4	1	3	S	curved	81.29
22	P12	3	2	1	S	concave	151.46
23	P12	3	2	2	M	simple	139.77
24	P12	3	2	3	L	curved	128.07
25	P13	3	3	1	L	concave	110.53
26	P13	3	3	2	S	simple	87.13
27	P13	3	3	3	M	curved	122.22
34	P14	4	3	1	L	simple	69.59
35	P14	4	3	2	S	concave	40.35
36	P14	4	3	3	M	curved	57.89
7	P15	1	3	1	L	curved	180.70
8	P15	1	3	2	S	simple	163.16
9	P15	1	3	3	M	concave	186.55

4 Analysis

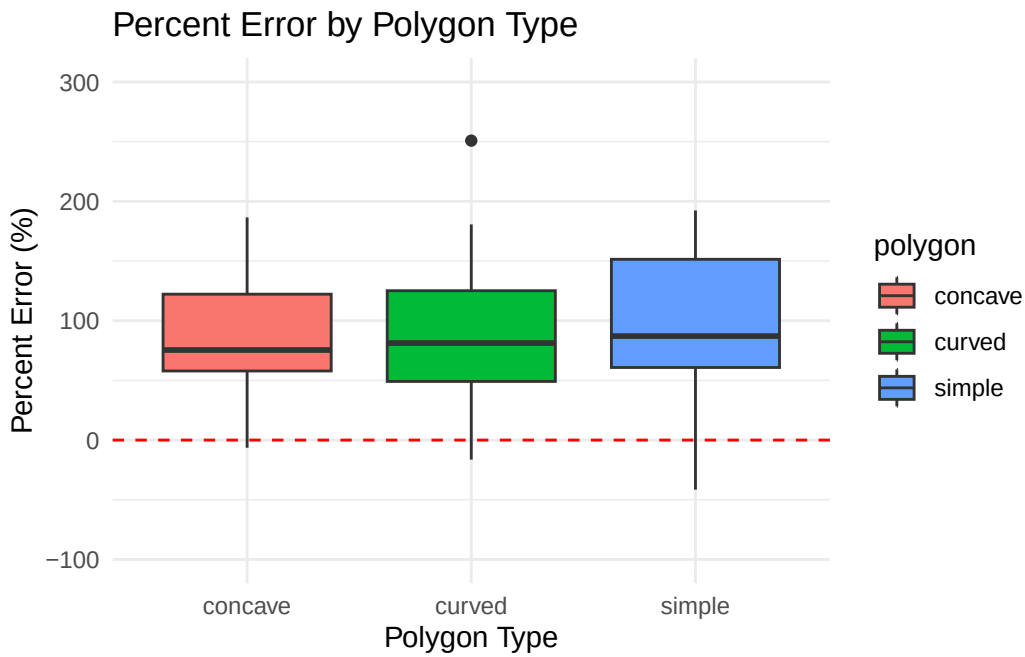
Exploratory Data:

Table 4: Summary statistics of percent error by scale treatment

scales	Mean	SD	Min	Max	N
L	89.98	79.55	-41.52	250.88	15
M	121.05	143.01	16.96	601.75	15
S	95.71	48.27	40.35	192.40	15



Note: 1 observation of 601% error on the medium scale was omitted for visual clarity



Note: 1 observation of 601% error on the curved polygon was omitted for visual clarity

The boxplots show that percent error is fairly similar across all three scale treatments (Large, Medium, Small), with most participants overestimating the polygon area by around 60–100%. There is one clear outlier with a percent error near 600%, which heavily influences the spread. Across polygon types, concave polygons had the most consistent estimates while curved polygons showed the most variation. Overall, the differences between groups appear modest, which we will be tested formally.

Hypothesis Testing: Test your hypothesis with an appropriate statistical test (e.g., ANOVA).

We test the following hypotheses for scale treatment:

$$H_0 : \tau_L = \tau_M = \tau_S = 0$$

$$H_a : \tau_i \neq 0 \text{ for at least one } i$$

and for polygon type:

$$H_0 : \beta_{\text{simple}} = \beta_{\text{curved}} = \beta_{\text{concave}} = 0$$
$$H_a : \beta_j \neq 0 \text{ for at least one } j$$

where τ_i and β_j are the fixed effects of scale and polygon type as defined in the model above, and Y_{ijk} is the signed percent error response. In context, H_0 for scale proposes that the geographic scale at which a polygon is displayed has no effect on the accuracy of area estimation. Equivalently, it proposes that participants estimate equally well regardless of whether the polygon occupies 40%, 10%, or 2% of the map frame. H_0 for polygon type asserts that polygon shape has no independent effect on estimation accuracy after controlling for scale and participant. While the participant blocking factor ρ_k is included to contain the individual variations in spatial reasoning ability, it's not the primary interest.

Tables, Figures, & Code:

4.0.1 Signed Percent Error

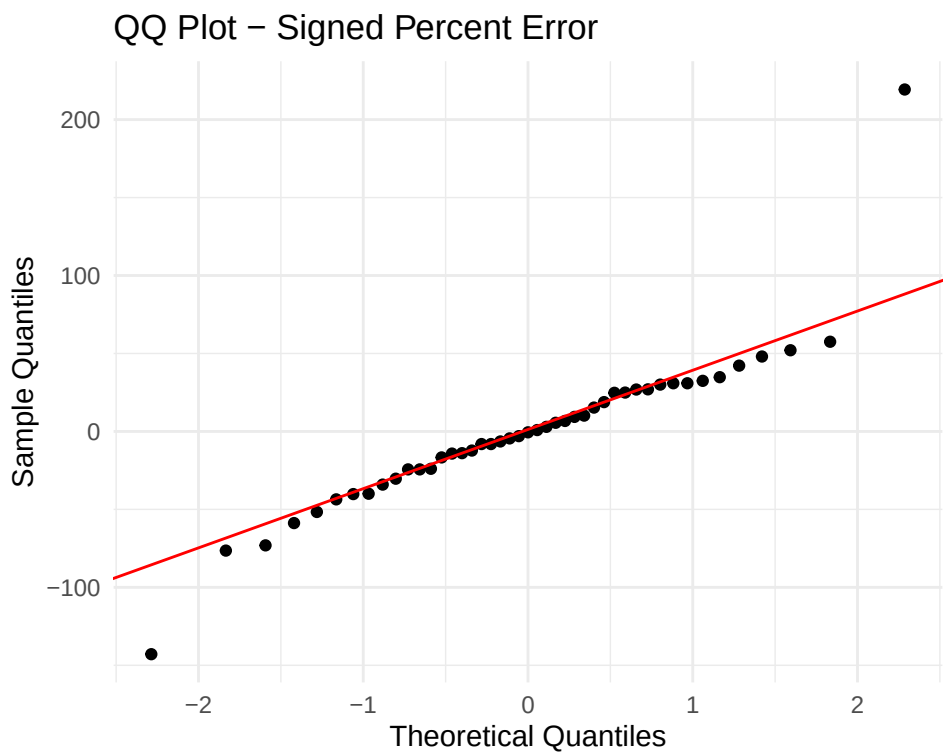
Table 5: ANOVA Table - Signed Percent Error

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	8202.471	4101.235	0.931	0.407
Polygon	2	5301.065	2650.533	0.602	0.555

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Participant	14	287714.480	20551.034	4.666	0.000
Residuals	26	114524.233	4404.778	NA	NA

Table 6: Treatment Parameter Estimates - Signed Percent Error

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	142.7940	43.1253	3.3111	0.0027
scalesM	31.0721	24.2344	1.2822	0.2111
scalesS	5.7310	24.2344	0.2365	0.8149
polygoncurved	-24.0546	24.2344	-0.9926	0.3301
polygonsimple	-21.8324	24.2344	-0.9009	0.3759



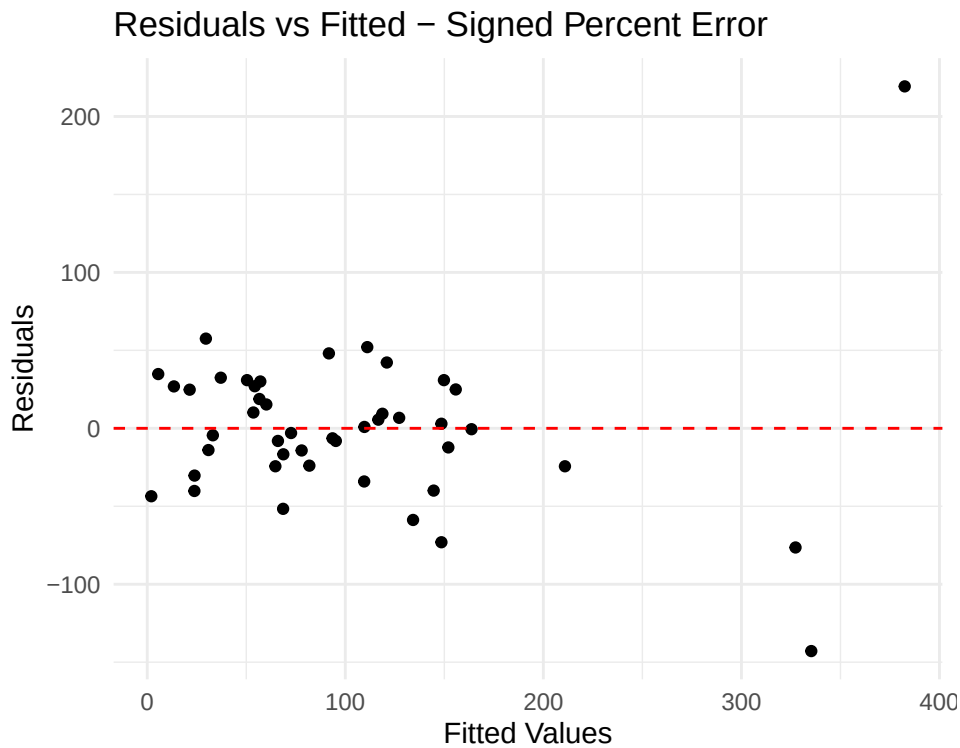


Table 7: Assumption Tests - Signed Percent Error

Test	Statistic	P.value	Result
Shapiro-Wilk	0.8614	0.0001	Fail
Levene (Scale)	0.1013	0.9039	Pass
Levene (Polygon)	0.3254	0.7240	Pass

Table 8: Detected Outliers (|Standardized Residual| > 2.5)

participant	scales	polygon	percent_error	std_resid
P7	M	concave	601.7544	3.304448

Through our exploratory data analysis, and now formally, by examining standardized residual values, we identified a major outlier in the data. Participant 7 grossly overestimated the area of the concave polygon when shown at a medium scale, with an estimate over 600% larger than the true value. With a standardized residual over 3, we decided to repeat analysis by omitting estimates from this participant. This decision was further supported by the mismatch we observed between the Q-Q plot and Shapiro-Wilk test of normality. The Q-Q plot indicated a pretty strong fit to a normal distribution, with the exclusion of one true point, and the Shapiro-Wilk test strongly indicated non-normality, invalidating our ANOVA results. If this outlying point had enough power to skew the Shapiro-Wilk test, its exclusion may reveal important insights in a revised ANOVA.

4.0.2 Signed Percent Error (Without Outlier)

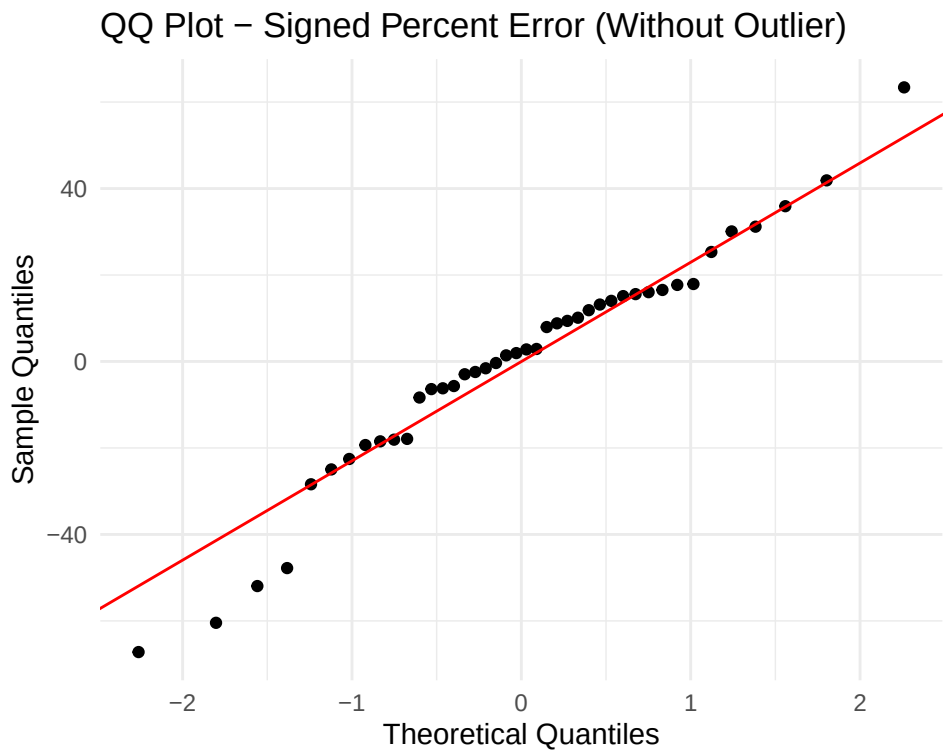
Table 9: ANOVA Table - Signed Percent Error (Without Outlier)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	833.126	416.563	0.348	0.710
Polygon	2	426.223	213.112	0.178	0.838
Participant	13	93048.712	7157.593	5.979	0.000

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Residuals	24	28732.558	1197.190	NA	NA

Table 10: Treatment Parameter Estimates - Signed Percent Error (Without Outlier)

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	131.8039	22.5737	5.8388	0.0000
scalesM	8.3221	13.1112	0.6347	0.5316
scalesS	10.8412	13.1112	0.8269	0.4165
polygoncurved	-1.3045	13.1112	-0.0995	0.9216
polygonsimple	6.0279	13.1112	0.4598	0.6498



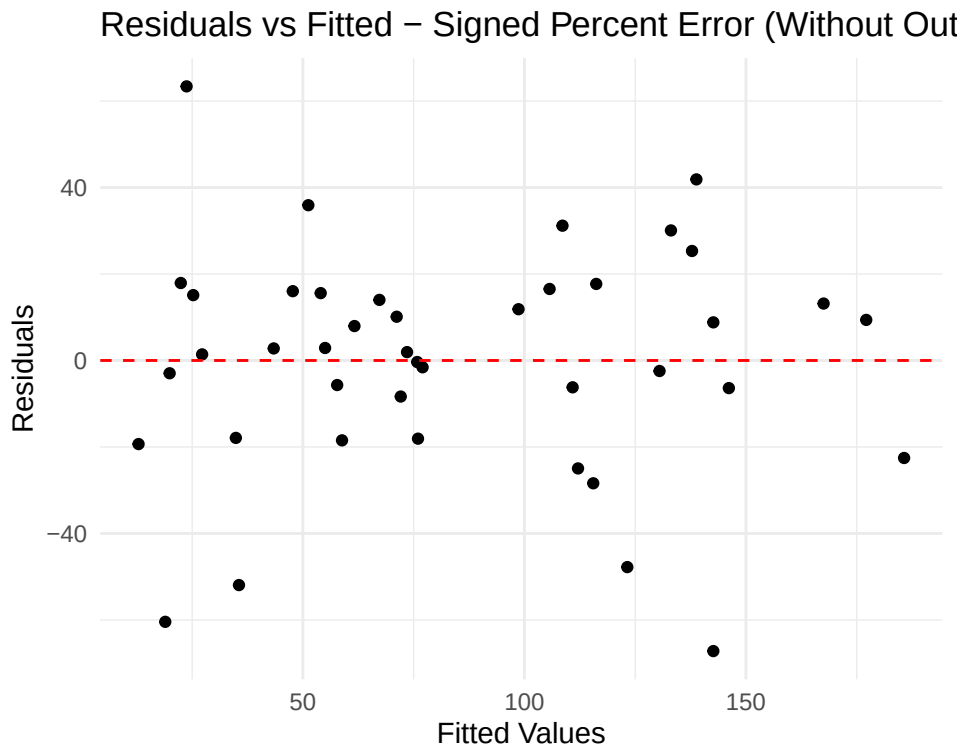


Table 11: Assumption Tests - Signed Percent Error (Without Outlier)

Test	Statistic	P.value	Result
Shapiro-Wilk	0.9615	0.1678	Pass
Levene (Scale)	0.5158	0.6011	Pass
Levene (Polygon)	0.0399	0.9609	Pass

The adjusted Shapiro-Wilk test confirms that with the exception of one outlying observation, the signed percentage error estimates were normally distributed and all model assumptions were satisfied. Shapiro-Wilk showed normality of residuals (W = 0.962, p = 0.168), and Levene's test showed constant variance across both scale (F = 0.516, p = 0.601) and polygon groups (F = 0.040, p = 0.961). However, the interpretation of the final ANOVA results did not change. Scale was still non-significant and polygon type was non-significant. The participant blocking factor remained highly significant (F = 5.979, p < 0.001), again confirming that individual differences in spatial reasoning ability dominate variation in estimation accuracy.

However, the need to omit valid observational data to perform a statistically valid analysis indicated that perhaps the signed percentage error was not the best metric for interpretation. Upon re-evaluation and research into literature, the use of a signed percentage error was revealed to be problematic. The most critical issue is the way the metric is bounded. Since the theoretical lowest value for a signed percentage error is -100%, where the participant makes an estimate of 0 square miles, the theoretical highest value was infinite. In practice, this meant our data was incredibly asymmetrical.

Furthermore, from the branch of psychophysics, there is an enormous wealth of peer-reviewed science that states that humans do not interpret area linearly, and as such, it would be erroneous to analyze area estimations with just a signed percentage error. Stevens's Power Law, first proposed back in 1957, states that the perceived magnitude of a stimulus is proportional to the magnitude of the stimuli raised to some experiment-derived constant n:

$$\psi = k \cdot \phi^n$$

While testing area estimates of a projected square at varying magnitudes, his work yielded an n of 0.7, which means that human perception of area is inherently non-linear (this has been supported by a number of preceding papers). By introducing a log-ratio transformation, we can derive a metric that has been empirically shown to best represent this phenomenon. We re-approached our analysis by defining our new response variable as:

$$\text{Log Ratio} = \log\left(\frac{\hat{a}}{a}\right)$$

where $\hat{a}$ is the participant's area estimate and $a = 1.71$ square miles is the true polygon area.

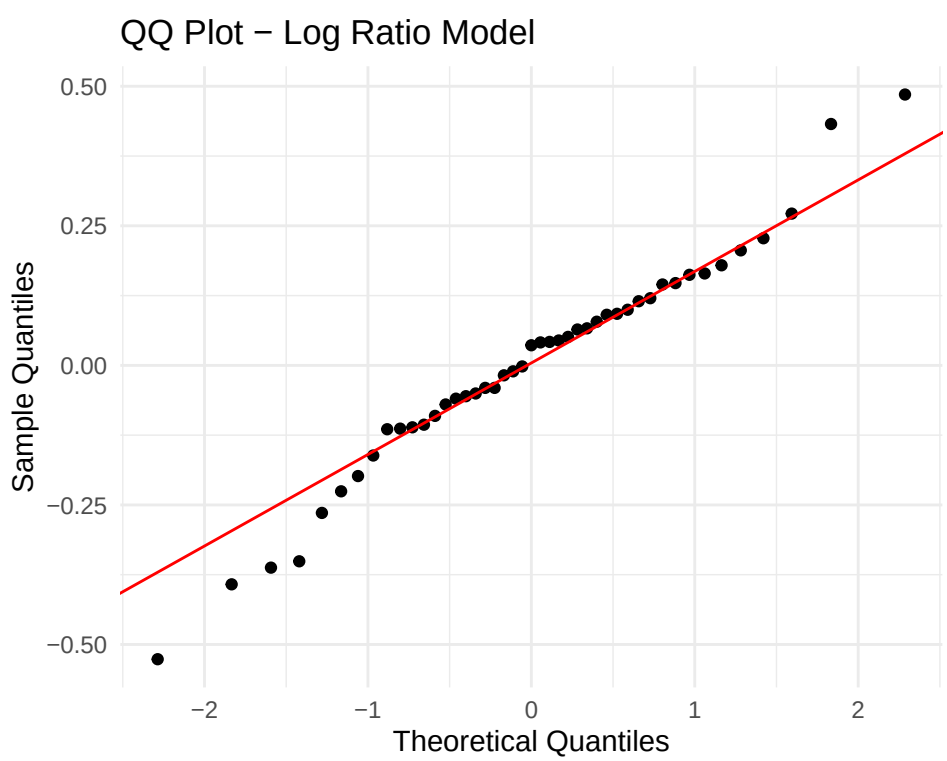
4.0.3 Log Ratio

Table 12: ANOVA Table - Log Ratio Model

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	0.146	0.073	1.089	0.351
Polygon	2	0.029	0.014	0.215	0.808
Participant	14	5.183	0.370	5.518	0.000
Residuals	26	1.744	0.067	NA	NA

Table 13: Treatment Parameter Estimates - Log Ratio Model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.8109	0.1683	4.8181	0.0001
scalesM	0.1335	0.0946	1.4114	0.1700
scalesS	0.1021	0.0946	1.0800	0.2901
polygoncurved	-0.0572	0.0946	-0.6051	0.5504
polygonsimple	-0.0494	0.0946	-0.5221	0.6060



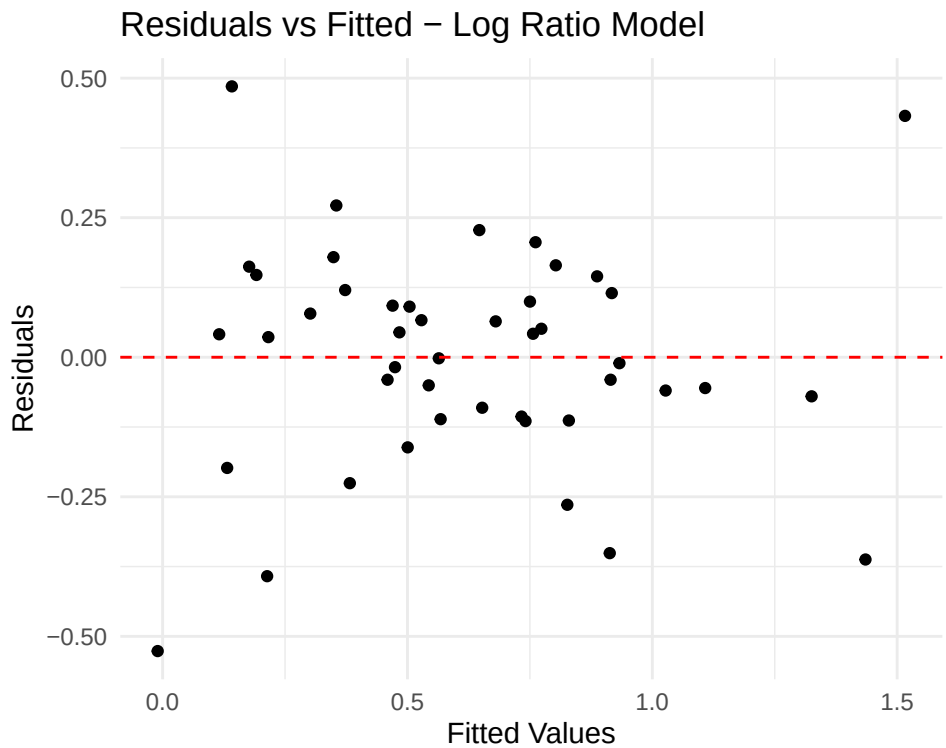


Table 14: Assumption Tests - Log Ratio Model

Test	Statistic	P.value	Result
Shapiro-Wilk	0.9747	0.4220	Pass
Levene (Scale)	0.1816	0.8346	Pass
Levene (Polygon)	0.8156	0.4492	Pass

```
No outliers detected under log ratio transformation.
```

Under the log ratio transformation of estimated areas, all model assumptions were satisfied without excluding data. Shapiro-Wilk showed normality of residuals ($W = 0.975$, $p = 0.422$), Levene's test indicated constant variance across scale ($F = 0.182$, $p = 0.835$) and polygon groups ($F = 0.816$, $p = 0.449$), and no observations passed the 2.5 standardized residual threshold for outlier detection. Scale was non-significant ($F = 1.089$, $p = 0.351$) and polygon type was non-significant ($F = 0.215$, $p = 0.808$). The participant blocking factor was highly significant ($F = 5.518$, $p < 0.001$), confirming that individual differences in spatial reasoning ability account for the dominant share of variation in estimation accuracy.

As a final check of data quality, we wanted to verify that the order in which participants viewed the maps did not affect their estimates. We had controlled for this in our aforementioned method for assigning treatments, but wanted additional validation. If our assumption of no learning effect was invalidated, we wanted to investigate what have occurred. This was performed by adding in the trial order as a covariate in the model, and then examining if the fitted coefficient was statistically significant.

Table 15: Model with Trial Order Covariate - Log Ratio

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	0.146	0.073	1.059	0.362
Polygon	2	0.029	0.014	0.209	0.813
Participant	14	5.183	0.370	5.365	0.000
Trial Order	1	0.020	0.020	0.283	0.600
Residuals	25	1.725	0.069	NA	NA

The model fit showed that trial order did not have a statistically significant effect on estimation accuracy, providing no evidence of systematic learning or fatigue across trials.

5 Conclusions

This experiment investigated whether the scale at which a polygon is displayed on a map affects a participant's ability to accurately estimate its area. We explored multiple transformations of the polygon area estimations and rigorously evaluated model assumptions and metrics to ensure statistically valid results. We found that the log ratio $\log(\hat{a}/a)$ transformation did the best job ensuring our data satisfied underlying assumptions and supporting scientific literature. In our 2-way ANOVA, we found that scale was non-significant ($F = 1.089$, $p = 0.351$) and polygon type was non-significant ($F = 0.215$, $p = 0.808$) at an alpha-level of 0.05. Variation in $\log(\hat{a}/a)$ was dominated by individual differences in spatial reasoning ability, and was very significant, with p-value less than 0.001. It was also evident in the ANOVA table, as the sum of squares from participant dwarfed our factors of interest. Consequently, we fail to reject both hypotheses. There is no sufficient evidence that large, medium, or small scale presentations nor polygon shape independently lead to statistically significant changes in area estimation accuracy. Parameter estimates further confirm this conclusion. Area estimates at medium scale were 0.134 log units higher than large scale ($SE = 0.095$, $p = 0.170$) and small scale estimates were 0.102 log units higher ($SE = 0.095$, $p = 0.290$), but neither were statistically significant. The intercept, of 0.811 ($SE = 0.168$, $p < 0.001$) was notably statistically significant. When converted back to signed estimation error, it indicates that the baseline participant estimated the polygon area to be approximately 2.25 larger than its true area. From this, we conclude that for our sampled population, participants consistently made egregious overestimates of polygon area, regardless of shape or scale.

5.1 Limitations and Improvements

There were several limitations that should be taken into account. First, many participants were unfamiliar with square miles as a unit, likely introducing systematic overestimation bias unrelated to scale. Future studies could provide unit conversion references or use more intuitive units. Second, the 30-second time limit may have prevented deliberate estimation, capturing first impressions rather than considered judgments. Third, the shared library setting created potential for participants to overhear each other's guesses, which may have anchored some responses. Finally, with only 15 participants, the study was incredibly underpowered to detect small effects of scale. While we collected 3 responses per participant to augment the size of data for analysis, it introduced a number of obstacles that made it difficult to properly control for variance across individuals. In this case, it cost us a whopping 14 degrees of freedom. Further work should record only one data point per participant, and should extend participants to be random effects rather than fixed effects to improve analysis and make results more generalizable.

6 References

Instituto Geográfico Militar de la República Argentina. (2009). POSGAR 2007 / Argentina 1 (EPSG:5343). EPSG Geodetic Parameter Registry. Retrieved from https://epsg.io/5343

Stevens, S. S. (1957). On the psychophysical law. Psychological Review, 64(3), 153–181. https://doi.org/10.1037/h0046162.

7 Appendices

- Slides for running trials: https://docs.google.com/presentation/d/1cQkHKFSfIFLclCvy6DIQ8lRVRss23OmEZ6OrBm-pAJ2I/edit?usp=sharing
- `R` code:

```r
#### Assigning Randomized + Balanced Treatments for Participants ####
library(agricolae)

scales <- c("L", "M", "S")
polygon_types <- c("curved", "simple", "concave")

# randomly assigning each participant to a specific replicate + row
set.seed(42)
```

```r
participant_assignment <- data.frame(
  replicate = rep(1:5, each = 3),
  row = rep(1:3, times = 5),
  # Randomly assigning participants through these slots
  participant = paste0("P", sample(1:15))
)

# Generating the 5 Latin Squares
design_df <- do.call(rbind, lapply(1:5, function(i) {
  book <- design.lsd(scales, seed = i * 10)$book
  book$replicate <- i

  # Assign polygon types to columns (1, 2, 3)
  set.seed(i)
  col_map <- setNames(sample(polygon_types), 1:3)
  book$polygon <- col_map[as.character(book$col)]

  return(book)
}))

# Merge on BOTH replicate and row to build final design dataframe
design_df <- merge(design_df, participant_assignment, by = c("replicate", "row"))

# 4. Sort neatly by participant (P1 to P15) and then by trial/column (1, 2, 3) to make implementing easy
design_df <- design_df[order(as.numeric(gsub("P", "", design_df$participant)), design_df$col),
                       c("participant", "replicate", "row", "col", "scales", "polygon")]

# Creating an empty response column
design_df$percent_error <- NA

cat("\nFull Design\n")
```

```
Full Design
```

```r
print(design_df)
```

```
   participant replicate row col scales polygon percent_error
43          P1         5   3   1      S  curved            NA
44          P1         5   3   2      M  simple            NA
45          P1         5   3   3      L concave            NA
31          P2         4   2   1      S  simple            NA
32          P2         4   2   2      M concave            NA
33          P2         4   2   3      L  curved            NA
19          P3         3   1   1      M concave            NA
20          P3         3   1   2      L  simple            NA
21          P3         3   1   3      S  curved            NA
25          P4         3   3   1      L concave            NA
26          P4         3   3   2      S  simple            NA
27          P4         3   3   3      M  curved            NA
4           P5         1   2   1      S  curved            NA
5           P5         1   2   2      M  simple            NA
6           P5         1   2   3      L concave            NA
40          P6         5   2   1      M  curved            NA
```

```
41          P6         5   2   2      L  simple            NA
42          P6         5   2   3      S concave            NA
28          P7         4   1   1      M  simple            NA
29          P7         4   1   2      L concave            NA
30          P7         4   1   3      S  curved            NA
13          P8         2   2   1      S  curved            NA
14          P8         2   2   2      M concave            NA
15          P8         2   2   3      L  simple            NA
16          P9         2   3   1      M  curved            NA
17          P9         2   3   2      L concave            NA
18          P9         2   3   3      S  simple            NA
22         P10         3   2   1      S concave            NA
23         P10         3   2   2      M  simple            NA
24         P10         3   2   3      L  curved            NA
7          P11         1   3   1      L  curved            NA
8          P11         1   3   2      S  simple            NA
9          P11         1   3   3      M concave            NA
1          P12         1   1   1      M  curved            NA
2          P12         1   1   2      L  simple            NA
3          P12         1   1   3      S concave            NA
37         P13         5   1   1      L  curved            NA
38         P13         5   1   2      S  simple            NA
39         P13         5   1   3      M concave            NA
34         P14         4   3   1      L  simple            NA
35         P14         4   3   2      S concave            NA
36         P14         4   3   3      M  curved            NA
10         P15         2   1   1      L  curved            NA
11         P15         2   1   2      S concave            NA
12         P15         2   1   3      M  simple            NA
```

```r
enter_data <- function(df, p_name, estimates) {

  # calculate the percent error based on the true area (1.71 sq miles)
  errors <- ((estimates - 1.71) / 1.71) * 100

  for(i in 1:3) {
    df$percent_error[df$participant == p_name & df$col == i] <- errors[i]
  }

  return(df)
}

#data collection
design_df <- enter_data(design_df, "P1", c(4.8, 4.5, 3.0))
design_df <- enter_data(design_df, "P2", c(3, 3, 3))
design_df <- enter_data(design_df, "P3", c(2, 1, 3.2))
design_df <- enter_data(design_df, "P4", c(3, 4, 4.5))
design_df <- enter_data(design_df, "P5", c(3.1, 2.8, 2.6))
design_df <- enter_data(design_df, "P6", c(2.4, 2.2, 1.6))
design_df <- enter_data(design_df, "P7", c(5, 12, 6))
design_df <- enter_data(design_df, "P8", c(2.4, 2.5, 2))
design_df <- enter_data(design_df, "P9", c(1.43, 2.8, 3.2))
design_df <- enter_data(design_df, "P10", c(3.2, 3.5, 4.1))
design_df <- enter_data(design_df, "P11", c(2.7, 2.9, 3.1))
design_df <- enter_data(design_df, "P12", c(4.3, 4.1, 3.9))
design_df <- enter_data(design_df, "P13", c(3.6, 3.2, 3.8))
```

```r
design_df <- enter_data(design_df, "P14", c(2.9, 2.4, 2.7))
design_df <- enter_data(design_df, "P15", c(4.8, 4.5, 4.9))

knitr::kable(design_df,
             caption = "Full experimental data with percent error by participant, polygon type, and scale
             digits = 2)
```

Table 16: Full experimental data with percent error by participant, polygon type, and scale

	participant	replicate	row	col	scales	polygon	percent_error
43	P1	5	3	1	S	curved	180.70
44	P1	5	3	2	M	simple	163.16
45	P1	5	3	3	L	concave	75.44
31	P2	4	2	1	S	simple	75.44
32	P2	4	2	2	M	concave	75.44
33	P2	4	2	3	L	curved	75.44
19	P3	3	1	1	M	concave	16.96
20	P3	3	1	2	L	simple	-41.52
21	P3	3	1	3	S	curved	87.13
25	P4	3	3	1	L	concave	75.44
26	P4	3	3	2	S	simple	133.92
27	P4	3	3	3	M	curved	163.16
4	P5	1	2	1	S	curved	81.29
5	P5	1	2	2	M	simple	63.74
6	P5	1	2	3	L	concave	52.05
40	P6	5	2	1	M	curved	40.35
41	P6	5	2	2	L	simple	28.65
42	P6	5	2	3	S	concave	-6.43
28	P7	4	1	1	M	simple	192.40
29	P7	4	1	2	L	concave	601.75
30	P7	4	1	3	S	curved	250.88
13	P8	2	2	1	S	curved	40.35
14	P8	2	2	2	M	concave	46.20
15	P8	2	2	3	L	simple	16.96
16	P9	2	3	1	M	curved	-16.37
17	P9	2	3	2	L	concave	63.74
18	P9	2	3	3	S	simple	87.13
22	P10	3	2	1	S	concave	87.13
23	P10	3	2	2	M	simple	104.68
24	P10	3	2	3	L	curved	139.77
7	P11	1	3	1	L	curved	57.89
8	P11	1	3	2	S	simple	69.59
9	P11	1	3	3	M	concave	81.29
1	P12	1	1	1	M	curved	151.46
2	P12	1	1	2	L	simple	139.77
3	P12	1	1	3	S	concave	128.07
37	P13	5	1	1	L	curved	110.53
38	P13	5	1	2	S	simple	87.13
39	P13	5	1	3	M	concave	122.22
34	P14	4	3	1	L	simple	69.59
35	P14	4	3	2	S	concave	40.35
36	P14	4	3	3	M	curved	57.89
10	P15	2	1	1	L	curved	180.70
11	P15	2	1	2	S	concave	163.16
12	P15	2	1	3	M	simple	186.55

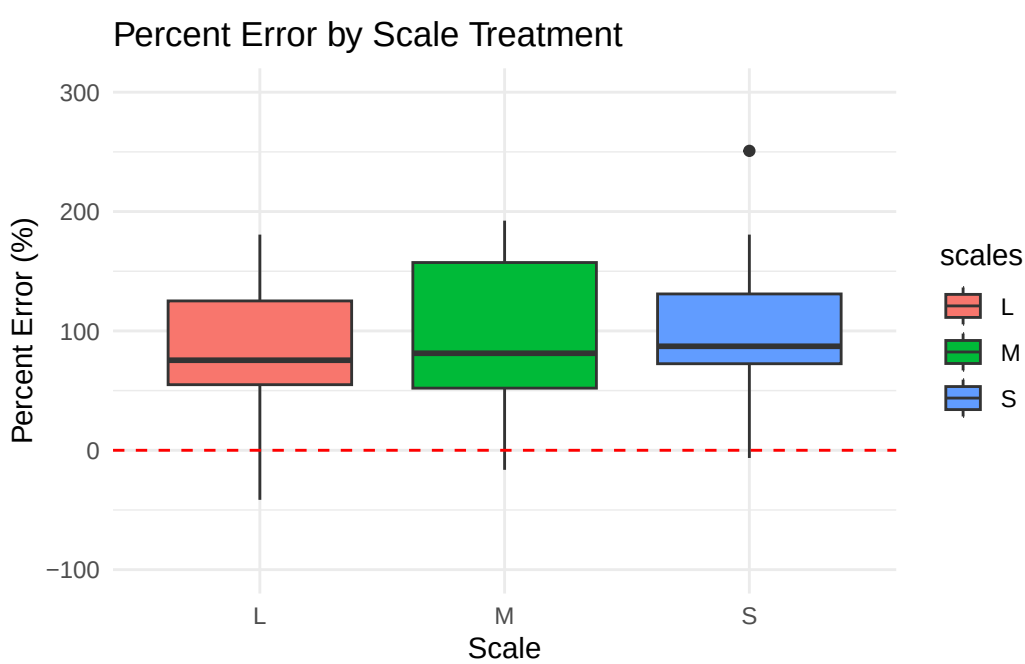
participant	replicate	row	col	scales	polygon	percent_error

```r
library(dplyr)
library(ggplot2)

# Basic summary statistics by scale
summary_stats <- design_df %>%
  group_by(scales) %>%
  summarise(
    Mean = round(mean(percent_error, na.rm = TRUE), 2),
    SD = round(sd(percent_error, na.rm = TRUE), 2),
    Min = round(min(percent_error, na.rm = TRUE), 2),
    Max = round(max(percent_error, na.rm = TRUE), 2),
    N = n()
  )

knitr::kable(summary_stats, caption = "Summary statistics of percent error by scale treatment")
```

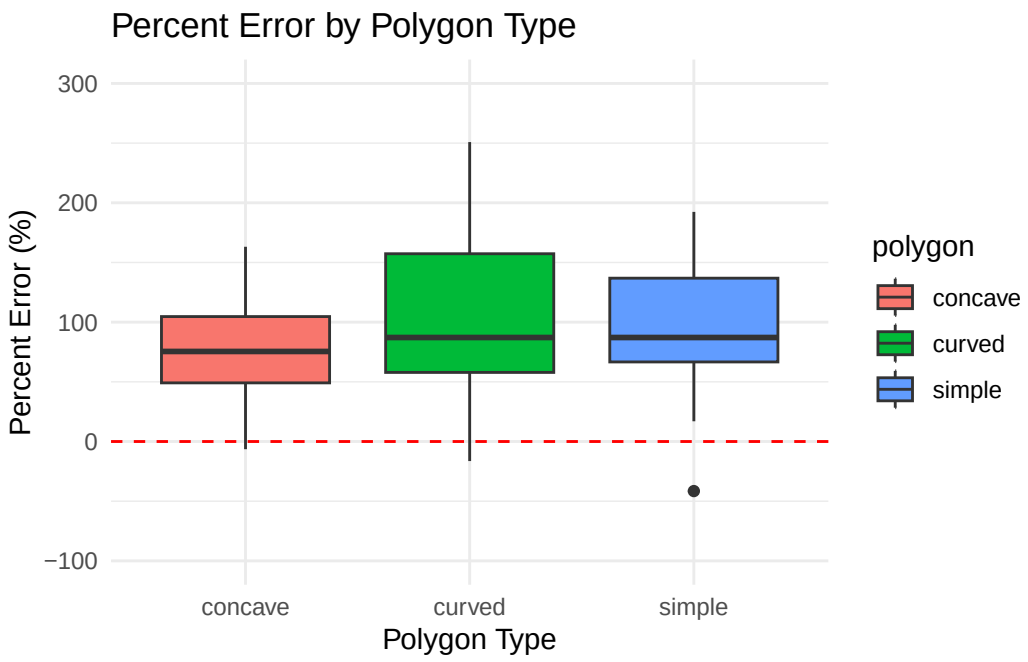
Table 17: Summary statistics of percent error by scale treatment

scales	Mean	SD	Min	Max	N
L	109.75	146.45	-41.52	601.75	15
M	96.61	64.33	-16.37	192.40	15
S	100.39	63.35	-6.43	250.88	15

```r
# Boxplot by Scale Treatment
ggplot(design_df, aes(x = scales, y = percent_error, fill = scales)) +
  geom_boxplot() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  coord_cartesian(ylim = c(-100, 300)) +
  labs(title = "Percent Error by Scale Treatment",
       x = "Scale", y = "Percent Error (%)") +
  theme_minimal()
```



```r
# Boxplot by polygon type
ggplot(design_df, aes(x = polygon, y = percent_error, fill = polygon)) +
  geom_boxplot() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  coord_cartesian(ylim = c(-100, 300)) +
  labs(title = "Percent Error by Polygon Type",
       x = "Polygon Type", y = "Percent Error (%)") +
  theme_minimal()
```




```r
# fit models and run diagnostics

# percent error - full data
mod_pct <- aov(percent_error ~ scales + polygon + participant, data = design_df)

sw_pct       <- shapiro.test(residuals(mod_pct))
design_df$resid_pct <- residuals(mod_pct)
lev_scale_pct   <- leveneTest(resid_pct ~ scales, data = design_df)
lev_poly_pct    <- leveneTest(resid_pct ~ polygon, data = design_df)
design_df$std_resid <- residuals(mod_pct) / sigma(mod_pct)
outliers <- design_df %>%
  filter(abs(std_resid) > 2.5) %>%
  select(participant, scales, polygon, percent_error, std_resid)

# percent error - P7 removed
df_noP7 <- design_df %>% filter(participant != "P7")
mod_noP7 <- aov(percent_error ~ scales + polygon + participant, data = df_noP7)

sw_noP7      <- shapiro.test(residuals(mod_noP7))
df_noP7$resid_noP7 <- residuals(mod_noP7)
lev_scale_noP7  <- leveneTest(resid_noP7 ~ scales, data = df_noP7)
lev_poly_noP7   <- leveneTest(resid_noP7 ~ polygon, data = df_noP7)

# log ratio - full data
design_df$log_ratio <- log(design_df$percent_error / 100 + 1)
mod_log <- aov(log_ratio ~ scales + polygon + participant, data = design_df)
```

```r
sw_log       <- shapiro.test(residuals(mod_log))
design_df$resid_log <- residuals(mod_log)
lev_scale_log   <- leveneTest(resid_log ~ scales, data = design_df)
lev_poly_log    <- leveneTest(resid_log ~ polygon, data = design_df)
design_df$std_resid_log <- residuals(mod_log) / sigma(mod_log)
outliers_log <- design_df %>%
  filter(abs(std_resid_log) > 2.5) %>%
  select(participant, scales, polygon, log_ratio, std_resid_log)
```

```r
pct_raw <- summary(mod_pct)[[1]]
pct_tbl <- data.frame(
  Df     = pct_raw$Df,
  SumSq  = round(pct_raw$`Sum Sq`, 3),
  MeanSq = round(pct_raw$`Mean Sq`, 3),
  Fvalue = round(pct_raw$`F value`, 3),
  Pvalue = round(pct_raw$`Pr(>F)`, 3),
  row.names = c("Scale", "Polygon", "Participant", "Residuals")
)
knitr::kable(pct_tbl,
             col.names = c("Df", "Sum Sq", "Mean Sq", "F value", "Pr(>F)"),
             caption = "ANOVA Table - Signed Percent Error",
             na_string = "-")
```

Table 18: ANOVA Table - Signed Percent Error

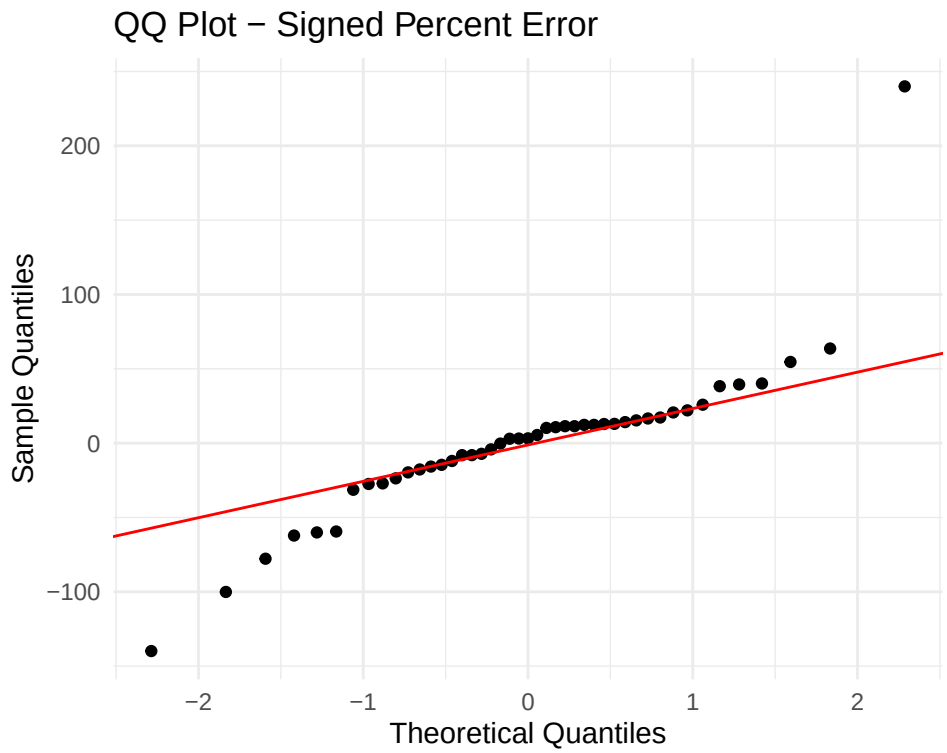
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	1372.335	686.167	0.144	0.867
Polygon	2	2465.777	1232.888	0.258	0.774
Participant	14	287714.480	20551.034	4.303	0.001
Residuals	26	124189.658	4776.525	NA	NA

```r
lm_pct <- summary.lm(mod_pct)$coefficients
lm_pct <- as.data.frame(lm_pct)
lm_pct <- lm_pct[grep("Intercept|scales|polygon", rownames(lm_pct)), ]
knitr::kable(round(lm_pct, 4),
             col.names = c("Estimate", "Std. Error", "t value", "Pr(>|t|)"),
             caption = "Treatment Parameter Estimates - Signed Percent Error")
```

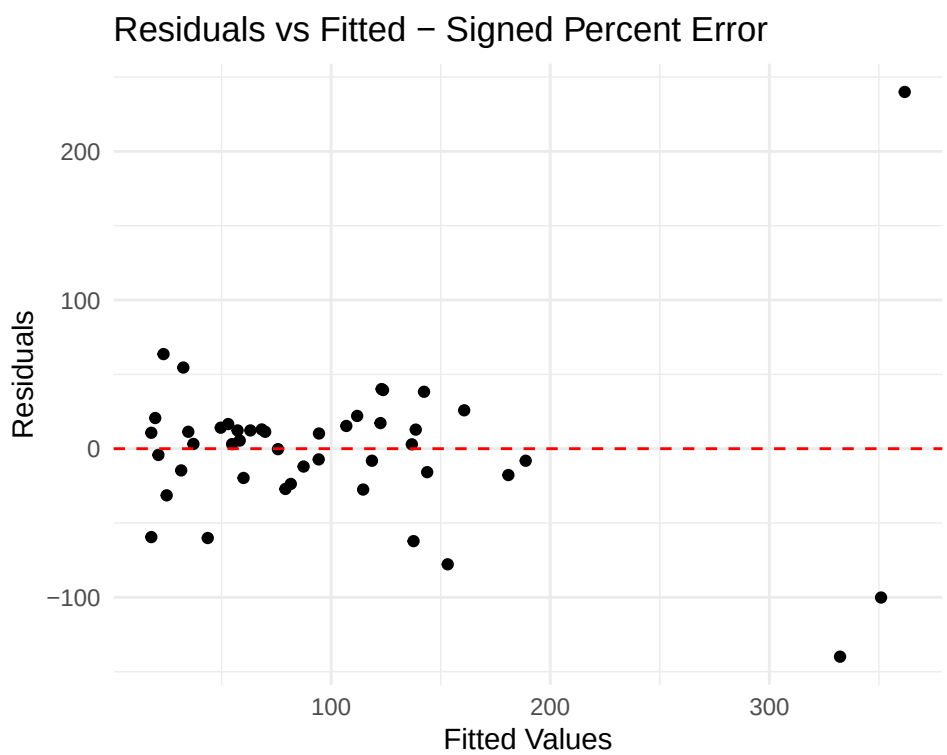
Table 19: Treatment Parameter Estimates - Signed Percent Error

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	153.2034	44.9083	3.4115	0.0021
scalesM	-13.1384	25.2363	-0.5206	0.6070
scalesS	-9.3567	25.2363	-0.3708	0.7138
polygoncurved	-1.4425	25.2363	-0.0572	0.9549
polygonsimple	-16.3743	25.2363	-0.6488	0.5221

```r
ggplot(data.frame(resid = residuals(mod_pct)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "QQ Plot - Signed Percent Error",
       x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal()
```




```r
ggplot(data.frame(fitted = fitted(mod_pct),
                  resid  = residuals(mod_pct)),
       aes(x = fitted, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted - Signed Percent Error",
       x = "Fitted Values", y = "Residuals") +
  theme_minimal()
```



```r
assump_pct <- data.frame(
  Test      = c("Shapiro-Wilk", "Levene (Scale)", "Levene (Polygon)"),
  Statistic = c(round(sw_pct$statistic, 4),
                round(lev_scale_pct[1,2], 4),
                round(lev_poly_pct[1,2], 4)),
  P.value   = c(round(sw_pct$p.value, 4),
                round(lev_scale_pct[1,3], 4),
                round(lev_poly_pct[1,3], 4)),
  Result    = c("Fail", "Pass", "Pass")
)
knitr::kable(assump_pct, caption = "Assumption Tests - Signed Percent Error")
```

Table 20: Assumption Tests - Signed Percent Error

Test	Statistic	P.value	Result
Shapiro-Wilk	0.8062	0.0000	Fail
Levene (Scale)	0.3774	0.6880	Pass
Levene (Polygon)	0.2241	0.8002	Pass

```r
knitr::kable(outliers, caption = "Detected Outliers (|Standardized Residual| > 2.5)")
```

Table 21: Detected Outliers (|Standardized Residual| > 2.5)

participant	scales	polygon	percent_error	std_resid
P7	L	concave	601.7544	3.472227

```r
noP7_raw <- summary(mod_noP7)[[1]]
noP7_tbl <- data.frame(
  Df     = noP7_raw$Df,
  SumSq  = round(noP7_raw$`Sum Sq`, 3),
  MeanSq = round(noP7_raw$`Mean Sq`, 3),
  Fvalue = round(noP7_raw$`F value`, 3),
  Pvalue = round(noP7_raw$`Pr(>F)`, 3),
  row.names = c("Scale", "Polygon", "Participant", "Residuals")
)
knitr::kable(noP7_tbl,
             col.names = c("Df", "Sum Sq", "Mean Sq", "F value", "Pr(>F)"),
             caption = "ANOVA Table - Signed Percent Error (Without Outlier)",
             na_string = "-")
```

Table 22: ANOVA Table - Signed Percent Error (Without Outlier)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	2128.274	1064.137	1.082	0.355
Polygon	2	4257.004	2128.502	2.164	0.137
Participant	13	93048.712	7157.593	7.277	0.000
Residuals	24	23606.629	983.610	NA	NA

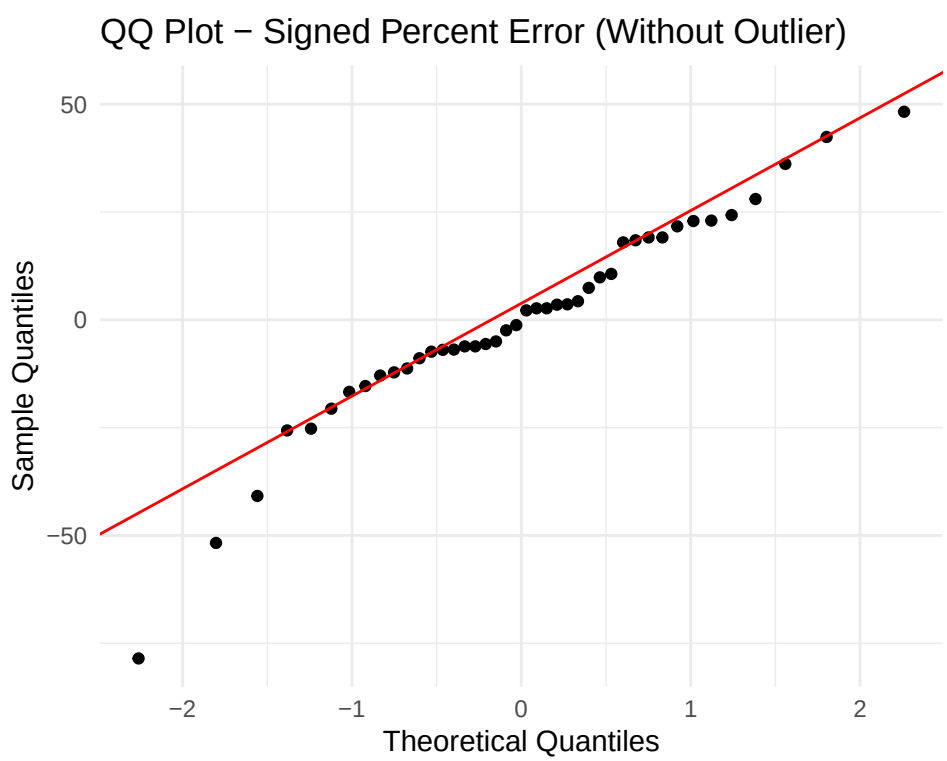
```r
lm_noP7 <- summary.lm(mod_noP7)$coefficients
lm_noP7 <- as.data.frame(lm_noP7)
lm_noP7 <- lm_noP7[grep("Intercept|scales|polygon", rownames(lm_noP7)), ]
```

```r
knitr::kable(round(lm_noP7, 4),
             col.names = c("Estimate", "Std. Error", "t value", "Pr(>|t|)"),
             caption = "Treatment Parameter Estimates - Signed Percent Error (Without Outlier)")
```

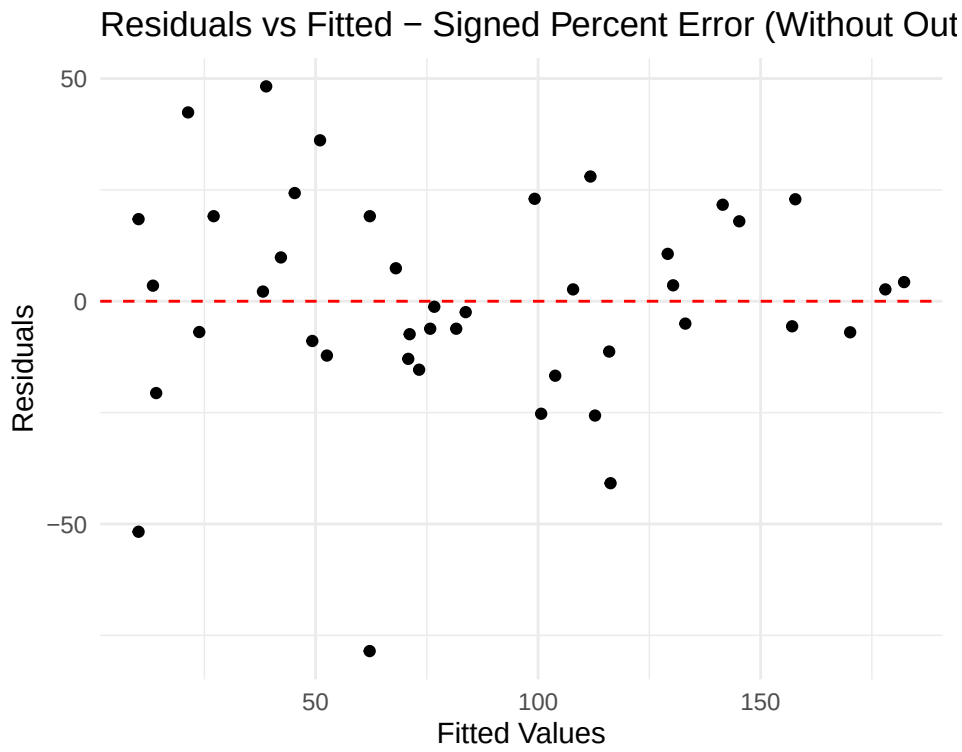
Table 23: Treatment Parameter Estimates - Signed Percent Error (Without Outlier)

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	116.2843	20.7063	5.6159	0.0000
scalesM	16.0804	11.8843	1.3531	0.1886
scalesS	16.8031	11.8843	1.4139	0.1702
polygoncurved	24.7173	11.8843	2.0798	0.0484
polygonsimple	12.8445	11.8843	1.0808	0.2905

```r
ggplot(data.frame(resid = residuals(mod_noP7)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "QQ Plot - Signed Percent Error (Without Outlier)",
       x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal()
```




```r
ggplot(data.frame(fitted = fitted(mod_noP7),
                  resid  = residuals(mod_noP7)),
       aes(x = fitted, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted - Signed Percent Error (Without Outlier)",
       x = "Fitted Values", y = "Residuals") +
  theme_minimal()
```




```r
assump_noP7 <- data.frame(
  Test      = c("Shapiro-Wilk", "Levene (Scale)", "Levene (Polygon)"),
  Statistic = c(round(sw_noP7$statistic, 4),
                round(lev_scale_noP7[1,2], 4),
                round(lev_poly_noP7[1,2], 4)),
  P.value   = c(round(sw_noP7$p.value, 4),
                round(lev_scale_noP7[1,3], 4),
                round(lev_poly_noP7[1,3], 4)),
  Result    = c("Pass", "Pass", "Pass")
)
knitr::kable(assump_noP7,
             caption = "Assumption Tests - Signed Percent Error (Without Outlier)")
```

Table 24: Assumption Tests - Signed Percent Error (Without Outlier)

Test	Statistic	P.value	Result
Shapiro-Wilk	0.9527	0.0809	Pass
Levene (Scale)	0.3358	0.7168	Pass
Levene (Polygon)	0.0460	0.9551	Pass

```r
log_raw <- summary(mod_log)[[1]]
log_tbl <- data.frame(
  Df     = log_raw$Df,
  SumSq  = round(log_raw$`Sum Sq`, 3),
  MeanSq = round(log_raw$`Mean Sq`, 3),
  Fvalue = round(log_raw$`F value`, 3),
  Pvalue = round(log_raw$`Pr(>F)`, 3),
  row.names = c("Scale", "Polygon", "Participant", "Residuals")
)
knitr::kable(log_tbl,
             col.names = c("Df", "Sum Sq", "Mean Sq", "F value", "Pr(>F)"),
```

```
             caption = "ANOVA Table - Log Ratio Model",
             na_string = "-")
```

Table 25: ANOVA Table - Log Ratio Model

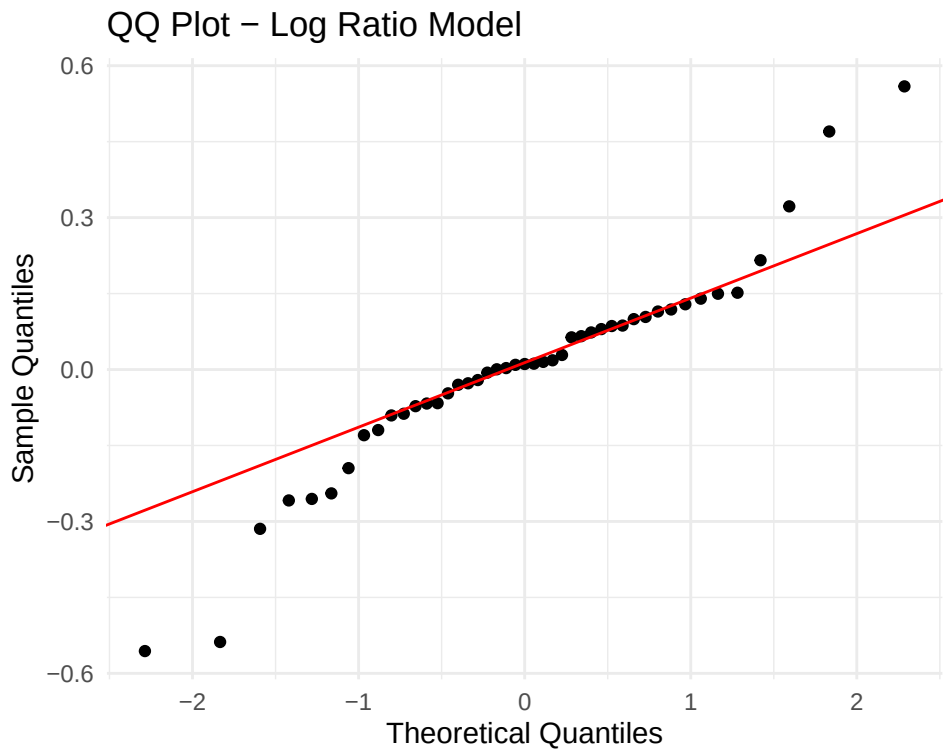
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	0.022	0.011	0.152	0.860
Polygon	2	0.055	0.027	0.386	0.684
Participant	14	5.183	0.370	5.222	0.000
Residuals	26	1.843	0.071	NA	NA

```
lm_log <- summary.lm(mod_log)$coefficients
lm_log <- as.data.frame(lm_log)
lm_log <- lm_log[grep("Intercept|scales|polygon", rownames(lm_log)), ]
knitr::kable(round(lm_log, 4),
             col.names = c("Estimate", "Std. Error", "t value", "Pr(>|t|)"),
             caption = "Treatment Parameter Estimates - Log Ratio Model")
```

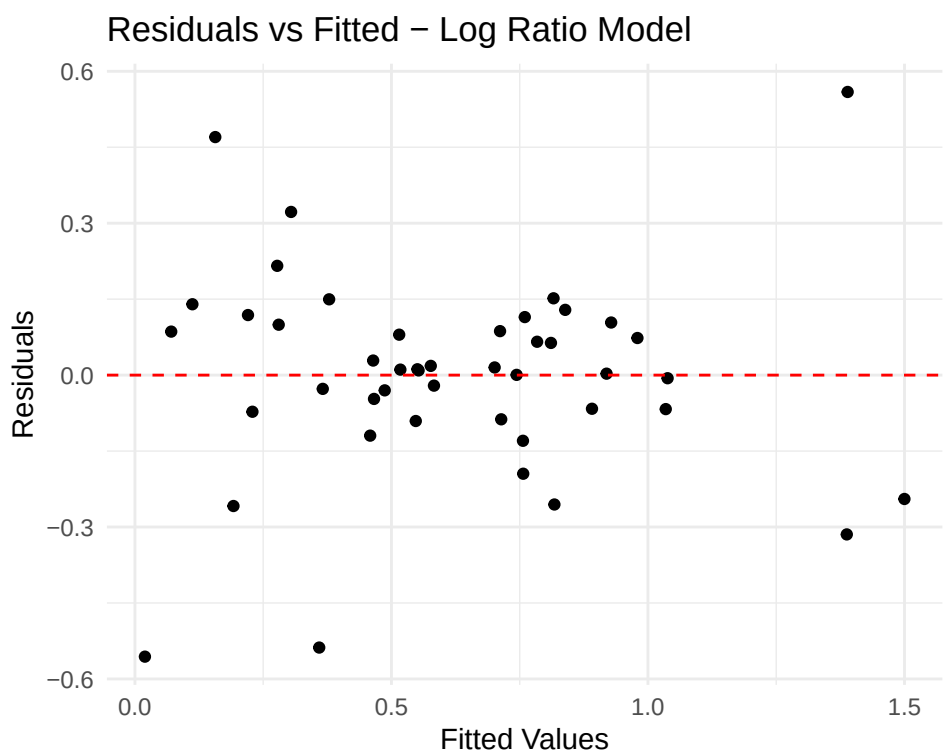
Table 26: Treatment Parameter Estimates - Log Ratio Model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.8176	0.1730	4.7259	0.0001
scalesM	0.0248	0.0972	0.2549	0.8008
scalesS	0.0535	0.0972	0.5506	0.5866
polygoncurved	0.0571	0.0972	0.5873	0.5621
polygonsimple	-0.0265	0.0972	-0.2723	0.7876

```
ggplot(data.frame(resid = residuals(mod_log)), aes(sample = resid)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "QQ Plot - Log Ratio Model",
       x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal()
```




```r
ggplot(data.frame(fitted = fitted(mod_log),
                  resid  = residuals(mod_log)),
       aes(x = fitted, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted - Log Ratio Model",
       x = "Fitted Values", y = "Residuals") +
  theme_minimal()
```



```r
assump_log <- data.frame(
  Test      = c("Shapiro-Wilk", "Levene (Scale)", "Levene (Polygon)"),
  Statistic = c(round(sw_log$statistic, 4),
                round(lev_scale_log[1,2], 4),
                round(lev_poly_log[1,2], 4)),
  P.value   = c(round(sw_log$p.value, 4),
                round(lev_scale_log[1,3], 4),
                round(lev_poly_log[1,3], 4)),
  Result    = c("Pass", "Pass", "Pass")
)
knitr::kable(assump_log, caption = "Assumption Tests - Log Ratio Model")
```

Table 27: Assumption Tests - Log Ratio Model

Test	Statistic	P.value	Result
Shapiro-Wilk	0.9319	0.0110	Pass
Levene (Scale)	0.5232	0.5964	Pass
Levene (Polygon)	0.0769	0.9261	Pass

```r
if(nrow(outliers_log) == 0){
  cat("No outliers detected under log ratio transformation.")
} else {
  knitr::kable(outliers_log, caption = "Outliers - Log Ratio Model")
}
```

```
No outliers detected under log ratio transformation.
```

```r
mod_order <- aov(log_ratio ~ scales + polygon + participant + as.numeric(col),
                 data = design_df)

order_raw <- summary(mod_order)[[1]]
order_tbl <- data.frame(
  Df     = order_raw$Df,
  SumSq  = round(order_raw$`Sum Sq`, 3),
  MeanSq = round(order_raw$`Mean Sq`, 3),
  Fvalue = round(order_raw$`F value`, 3),
  Pvalue = round(order_raw$`Pr(>F)`, 3),
  row.names = c("Scale", "Polygon", "Participant", "Trial Order", "Residuals")
)
knitr::kable(order_tbl,
             col.names = c("Df", "Sum Sq", "Mean Sq", "F value", "Pr(>F)"),
             caption = "Model with Trial Order Covariate - Log Ratio",
             na_string = "-")
```

Table 28: Model with Trial Order Covariate - Log Ratio

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scale	2	0.022	0.011	0.149	0.862
Polygon	2	0.055	0.027	0.380	0.688
Participant	14	5.183	0.370	5.136	0.000
Trial Order	1	0.041	0.041	0.575	0.455
Residuals	25	1.802	0.072	NA	NA